

Tao 6

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1 Problems

1. Let $(x_n) \in \mathbb{R}^\omega$ be nonincreasing and bounded below. Let $L = \inf\{x_n\}$. Then $(x_n) \rightarrow L$.

Let $\epsilon > 0$. Consider $x_N \in [L, L + \epsilon) \cap \{x_n\} \neq \emptyset$ (for otherwise $L + \epsilon$ would be a lower bound of $\{x_n\}$). For $n \geq N$, $L \leq x_n \leq x_N < L + \epsilon$, so $\{x_n\}_{n \geq N} \subset [L, L + \epsilon)$, ie $(x_n) \rightarrow L$.

2. Let $(x_n) \in \mathbb{R}^\omega$ be Cauchy and (x_{a_n}) a subsequence. Then $(x_n), (x_{a_n})$ are equivalent.

Let $\epsilon > 0$. Pick N st $n, m \geq N \rightarrow |x_n - x_m| < \epsilon$. In particular, $a_n \geq n$, so $|x_{a_n} - x_n| < \epsilon$ for $n \geq N$.

3. Every Cauchy sequence in $\mathbb{R}$ is equivalent to a sequence in $\mathbb{Q}$.

Pick $q_n \in (x_n, x_n + 1/n)$. Then $(q_n) \in \mathbb{Q}^\omega$. (q_n) is Cauchy, so $(q_n) \in \mathbb{R}$.

4. $\mathbb{R}$ is complete.

Let $x \in \mathbb{R}^\omega$ be Cauchy. Pick an equivalent sequence $x = (x_n) \in \mathbb{Q}^\omega$, so $x \in \mathbb{R}$. Let $T = \{n \in N : x_n > x\}$, $B = \{n \in N : x_n \leq x\}$ partition N (implicit application of trichotomy of $\mathbb{R}$). WLOG T is infinite. Suppose FTSOC $x(T)$ has no upper bound. Construct $x_{a_1} < x_{a_2} < x_{a_3} < \dots$ with $a_1 < a_2 < a_3 < \dots$ ($1, 2, 3, \dots, a_n$ is finite), so $x_{a_n} \geq x_{a_1}$ and $(x_{a_n}) \geq x_{a_1} > x$. Since $(x_{a_n}), (x_n)$ are equivalent, $x = (x_{a_n})$, so $x > x$, cont.

Pick $x_{a_1} \in \{x_n\}$ maximal. Pick x_{a_n} to be the maximum of $\{x_m : m > a_n, m \in T\}$, possible since T is infinite. Then $x_{a_1} \geq x_{a_2} \geq x_{a_3} \geq \dots$. Let $L = \inf\{x_{a_n}\}$ (since $x_{a_n} > x$ is bounded). Then $(x_{a_n}) \rightarrow L$.

$L \leq x_{a_n} \rightarrow L \leq (x_{a_n}) = (x_n) = x$. $L \geq x$ (since x is a lower bound of $x(T)$), so $L = x$. $(x_{a_n}) \rightarrow x$, so $(x_n) \rightarrow x$, AWD.

Let $(X_1, \rho_1), (X_2, \rho_2)$ be metric spaces. Then $(X_1 \times X_2, \rho_1 + \rho_2)$ gives the product topology.

Let (X, ρ) be the (usual) discrete metric space. Then every Cauchy (x_n) is eventually constant and converges to an element of X .

Let $K \subset \mathbb{R}$ have the subspace topology (inherits discrete topology) and the usual metric. Let $x_n = 1/n$ be Cauchy in K , and therefore $\mathbb{R}$. Then $(x_n) \rightarrow 0$ in $\mathbb{R}$, and this limit is unique, so K is not complete. Completeness is not a purely topological property (depends on metric).

$(X, \rho), (Y, \rho')$ are complete iff $(X \times Y, \rho + \rho')$ is complete. Let $((x_n, y_n))$ be Cauchy. Then $(x_n), (y_n)$ are Cauchy, so $x_n \rightarrow x, y_n \rightarrow y$. Let $B_\rho(x, \epsilon) \times B_{\rho'}(y, \epsilon')$ be a basis containing (x, y) . Pick N such that $\{x_n\}_{n \geq N} \subset B_\rho(x, \epsilon), \{y_n\}_{n \geq N} \subset B_{\rho'}(y, \epsilon')$, so $\{(x_n, y_n)\}_{n \geq N} \subset B_\rho(x, \epsilon) \times B_{\rho'}(y, \epsilon')$.

Suppose $(X \times Y, \rho + \rho')$ is complete. Let (x_n) be Cauchy, so (x_n, y) is Cauchy for some fixed y . Let $(x_n, y) \rightarrow (x, y)$. Then $x_n \rightarrow x$, so (X, ρ) is complete.

Extend to countable spaces?

Let (Y, ρ) be complete. Then $\mathcal{C}(X, Y)$ endowed with the metric $\bar{\rho}$ is complete.

Suppose (f_n) of continuous functions is Cauchy. Let $1 > \epsilon > 0$ and pick N such that $n, m \geq N \rightarrow \bar{\rho}(f_n, f_m) < \epsilon/2$, so $\rho(f_n(x), f_m(x)) < \epsilon/2$ for all $x \in X$. Hence, $(f_n(x))$ is Cauchy in ρ , so by completeness let $f_n(x) \rightarrow f(x)$.

$(f_n(x))_{n \geq N}$ is a subsequence of $(f_n(x))$, so $f_n(x)_{n \geq N} \rightarrow f(x)$, so

$$f(x) \in \overline{\{f_n(x)\}_{n \geq N}} \subset \overline{B(f_m(x), \epsilon/2)} \subset \bar{B}(f_m(x), \epsilon/2).$$

That is, $\rho(f(x), f_m(x)) \leq \epsilon/2$, so for $m \geq N$, $\bar{\rho}(f(x), f_m(x)) \leq \epsilon/2 < \epsilon$, so (f_n) converges uniformly to f .

Let $x_0 \in X$ and $V = B(f(x_0), \epsilon)$ with $\epsilon < 1$ be open. Pick N such that $\bar{\rho}(f_N, f) \leq \epsilon/4$, so $\rho(f_N(x), f(x)) < \epsilon/3$ for all $x \in X$. Since f_N is continuous, pick $U(x_0)$ such that $f_N(U) \subset B(f_N(x_0), \epsilon/3)$.

Then if $x \in U$,

$$\begin{aligned} \rho(f(x), f(x_0)) &\leq \rho(f(x), f_N(x)) + \rho(f_N(x), f_N(x_0)) + \rho(f_N(x_0), f(x_0)) \\ &< \epsilon/3 + \epsilon/3 + \epsilon/3 = \epsilon. \end{aligned}$$

Hence, f is continuous.

Let ρ, ρ' be metrics on X such that (X, ρ) is complete and there exist fixed $\alpha, \beta > 0$ such that $\alpha\rho(x, y) \leq \rho'(x, y) \leq \beta\rho(x, y)$ for all $x, y \in X$. Then (X, ρ')

is complete.

Let (x_n) be Cauchy in ρ' . Let $\epsilon > 0$ and pick N such that $n, m \geq N \rightarrow \rho'(x_n, x_m) < \alpha\epsilon$. Then $\rho(x_n, x_m) \leq \frac{1}{\alpha}\rho'(x_n, x_m) < \alpha\epsilon/\alpha = \epsilon$, so (x_n) is Cauchy in ρ . Let $x_n \rightarrow x$ in ρ . I claim $B_\rho(x, \epsilon/\beta) \subset B_{\rho'}(x, \epsilon)$, from which it follows $x_n \rightarrow x$ in ρ' . Let $\rho(x, y) < \epsilon/\beta$. Then $\rho'(x, y) \leq \beta\rho(x, y) < \epsilon$. That is, $y \in B_{\rho'}(x, \epsilon)$.

Corollary: let $\rho(x, y) = \sum |x_i - y_i|$, $d(x, y) = \sqrt{\sum (x_i - y_i)^2}$. By Cauchy (and others),

$$\frac{1}{n} \left(\sum |x_i - y_i| \right)^2 \leq \sum (x_i - y_i)^2 \leq \left(\sum |x_i - y_i| \right)^2,$$

ie $\frac{1}{\sqrt{n}}\rho(x, y) \leq d(x, y) \leq \rho(x, y)$, so $(\mathbb{R}^n, d)$ is complete.

If (Y, ρ) is complete, so is $(Y^X, \bar{\rho})$ (first part of earlier proof).

$\mathbb{R}$ is complete. Let (x_n) be Cauchy in $(\mathbb{R}, d)$. Then (x_n) is bounded, so $\{x_n\} \subset [-M, M]$. Let (x_{a_n}) be a convergent subsequence. Then $(\mathbb{R}, d)$ is complete.